

①

Each teacher has 4 options

① a

8 teachers can be assigned to 4 schools in $\underbrace{4 \times 4 \times \dots 4}_{8 \text{ times}} = 4^8$

8 teachers partitioned across 4 schools st
each school gets 2 teachers

$$= \frac{8!}{2! 2! 2! 2!}$$

ways such that first school
gets exactly 4 teachers assigned $= \binom{8}{4} \times 3 \times 3 \times 3 \times 3$

Prob that first school gets exactly
4 teachers assigned

$$= \frac{\binom{8}{4} \times 3^4}{4^8}$$

① b

Let x_i be # of blackboards given to i -th school.

of (x_1, x_2, x_3, x_4) st $x_1 + x_2 + x_3 + x_4 = 8$.

where x_i 's are non-negative integers

$\Rightarrow x_i \geq 0$.

Equivalent to writing $8+4-1=11$ dashes and choosing

— | — — | — — | — — —

3 bars out of 11.

$$\Rightarrow x_1=1, x_2=2=x_3, x_4=3.$$

$\Rightarrow$ # of ways to distribute / divide ^{identical} 8 boards among 4 schools
 $= \boxed{\binom{11}{3}}.$

$\Rightarrow$ # of ways to distribute st each school gets at least 1 board.

$$x_1 + x_2 + x_3 + x_4 = 8$$

$$\text{st } x_i \geq 1.$$

$$\text{let } x_i = \hat{x}_i + 1$$

$$\Rightarrow \hat{x}_1 + \hat{x}_2 + \hat{x}_3 + \hat{x}_4 = 4.$$

$$\hat{x}_i \geq 0.$$

$$\# \text{ of possibilities} = \binom{4+4-1}{4-1} = \boxed{\binom{7}{3}}$$

Prob that 1st school gets exactly 4 boards

$$x_1 + x_2 + x_3 + x_4 = 8$$

$$\text{if } x_1 = 4 \Rightarrow x_2 + x_3 + x_4 = 4.$$

$$\# \text{ of possibilities} = \binom{4+3-1}{3-1} = \boxed{\binom{6}{2}}$$

$$\textcircled{2} \cdot \textcircled{a} \cdot \mathcal{A} = [-1, 2].$$

$$A = \{x \in \mathcal{A} \mid x < 0\} = [-1, 0)$$

$$B = \{x \in \mathcal{A} \mid |x - 0.5| < 0.5\} = (0, 1)$$

$$C = \{x \in \mathcal{A} \mid x > 0.75\} = (0.75, 2].$$

$$\textcircled{a} \quad P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(\emptyset)}{P(B)} = 0.$$

$$\textcircled{b} \quad P(B|C) = \frac{P(B \cap C)}{P(C)} = \frac{P((0.75, 1))}{P((0.75, 2])}.$$

$$= \frac{\left(\frac{0.25}{3}\right)}{\left(\frac{1.25}{3}\right)} = \frac{1}{5} = 0.2.$$

$$\textcircled{c} \quad P(A|C^c) = \frac{P(A \cap C^c)}{P(C^c)} = \frac{P([-1, 0))}{P([-1, 0.75])}.$$

$$= \frac{\frac{1}{3}}{\frac{1.75}{3}} = \frac{1}{1.75} = \frac{4}{7}$$

$$\textcircled{d} \quad P(B|C^c) = \frac{P(B \cap C^c)}{P(C^c)} = \frac{P\left(\overset{3}{(0, 0.75]}\right)}{P([-1, 0.75])} = \frac{0.75}{1.75}.$$

$$= 3/7.$$

(b)

(a) $\frac{1}{1+e^{-x}}$ as $x \uparrow$, $e^{-x} \downarrow$, $\frac{1}{1+e^{-x}} \uparrow$

monotonically increasing ✓

$$\lim_{x \rightarrow -\infty} \frac{1}{1+e^{-x}} = 0 \quad \checkmark$$

$$\lim_{x \rightarrow \infty} \frac{1}{1+e^{-x}} = 1 \quad \checkmark$$

Yes, a valid cdf.

Continuous function as well ✓

(b)

$1 - e^{-x}$ as $x \uparrow$, $e^{-x} \downarrow$, $1 - e^{-x} \uparrow$

monotonically increasing ✓

$$\lim_{x \rightarrow -\infty} 1 - e^{-x} = -\infty \neq 0 \quad \times$$

Not valid

©

$$\frac{e^x}{e^x + e^{-x}} = \frac{1}{1 + e^{-2x}}$$

$x \uparrow$

$e^{2x} \downarrow$ $1 + e^{-2x} \uparrow$

Similar to part ©

Yes, valid CDF

©

$\mathcal{F}, \mathcal{G}$

(i) $\mathcal{F} \cap \mathcal{G}$

union: $A \in \mathcal{F} \cap \mathcal{G}, B \in \mathcal{F} \cap \mathcal{G}$

closure:

$\Rightarrow A \in \mathcal{F}, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$

$A \in \mathcal{G}, B \in \mathcal{G} \Rightarrow A \cup B \in \mathcal{G}$

as $\mathcal{F}, \mathcal{G}$ are event space.

$\left. \begin{array}{l} A \cup B \in \mathcal{F} \\ A \cup B \in \mathcal{G} \end{array} \right\} \Rightarrow A \cup B \in \mathcal{F} \cap \mathcal{G}$

Compliment $A \in \mathcal{F} \cap \mathcal{G} \Rightarrow A \in \mathcal{F}, A \in \mathcal{G}$
 $\Rightarrow A^c \in \mathcal{F}, A^c \in \mathcal{G}$
 $\Rightarrow A^c \in \mathcal{F} \cap \mathcal{G}$

Yes $\mathcal{F} \cap \mathcal{G}$ is an event space as $\phi \in \mathcal{F}, \phi \in \mathcal{G}$.

(ii) $\mathcal{F} \cup \mathcal{G}$. $\Omega = \{1, 2, 3, 4, 5, 6\}$

$\mathcal{F} = \{\phi, \Omega, \{1\}, \{2, 3, 4, 5, 6\}\}$

$\mathcal{G} = \{\phi, \Omega, \{2\}, \{1, 3, 4, 5, 6\}\}$

$\mathcal{F} \cup \mathcal{G} = \{\phi, \Omega, \{1\}, \{2\}, \{2, 3, 4, 5, 6\}, \{1, 3, 4, 5, 6\}\}$

In the above example

$\mathcal{F}, \mathcal{G}$ are event spaces but $\mathcal{F} \cup \mathcal{G}$ is not.

$$\begin{aligned}
 \textcircled{3} \textcircled{a} P(R=S) &= \underbrace{P(R=S | S=0)}_{1-\epsilon_0} \underbrace{P(S=0)}_p + \underbrace{P(R=S | S=1)}_{1-\epsilon_1} \underbrace{P(S=1)}_{1-p} \\
 &= (1-\epsilon_0)p + (1-\epsilon_1)(1-p) \\
 &= 1 - (\epsilon_0 p + \epsilon_1(1-p)).
 \end{aligned}$$

$$\textcircled{b} \text{ Let } R_i = S_i + \underbrace{E_i}_{\text{Error R.V.}}. \quad E_i | S_i=0 = \begin{cases} 0 & \text{w.p. } 1-\epsilon_0 \\ 1 & \text{w.p. } \epsilon_0 \end{cases}$$

$$P(R_1 R_2 R_3 R_4 = 1011 | S_1 S_2 S_3 S_4 = 1011) = \begin{cases} 0 & \text{w.p. } 1-\epsilon_1 \\ 1 & \text{w.p. } \epsilon_1 \end{cases}$$

$$= P(E_1 E_2 E_3 E_4 = 0000 | S_1 S_2 S_3 S_4 = 1011)$$

$$= (1-\epsilon_1)^3 (1-\epsilon_0).$$

©

Suppose we want to send 0,
we transmit 000. This can be received as
000, 001, 100, 001, 110, 101, 011 or 111.
depending on the error in the channel.

Using the decoding scheme 0 is correctly recovered
when 0's are in majority i.e., when the received
vector is one of 000, 001, 100 or 001.

$$\text{let } E = \{000, 001, 100, 001\}$$

$$\begin{aligned} \text{Prob of correct decoding} &= P(\bigcup_{i=1}^4 E_i \mid S_1 S_2 S_3 = 000) \\ &= (1 - \epsilon_0)^3 + 3(1 - \epsilon_0)^2 \epsilon_0. \end{aligned}$$

④

$$P(S_1 S_2 S_3 = 000 \mid R_1 R_2 R_3 = 101).$$

$$= \frac{P(R_1 R_2 R_3 = 101 \mid S_1 S_2 S_3 = 000)}{P(R_1 R_2 R_3 = 101)}.$$

$$= \frac{P(R_1 R_2 R_3 = \underline{101} \mid S_1 S_2 S_3 = \underline{000}) P(S_1 S_2 S_3 = 000)}{P(R_1 R_2 R_3 = 101 \mid S_1 S_2 S_3 = 000) P(S_1 S_2 S_3 = 000) + P(R_1 R_2 R_3 = 101 \mid S_1 S_2 S_3 = 111) P(S_1 S_2 S_3 = 111)}$$

100 errors

$$P(S_1 S_2 S_3 = 000) = P(S=0) = p.$$

$$= \frac{(1-\epsilon_0) \epsilon_0^2 \times p}{(1-\epsilon_0) \epsilon_0^2 p + (1-\epsilon_1)^2 (\epsilon_1) (1-p)}.$$

② Prob of correct decoding on sending 000 = $(1-\epsilon_0)^3 + 3(1-\epsilon_0)^2 \epsilon_0$ } = e_1

Prob of correct decoding on sending 0 = $1-\epsilon_0$ } = e_0

Improvement is possible if
 $e_1 - e_0 \geq 0$.

$$\Leftrightarrow (1-\epsilon_0)^3 + 3(1-\epsilon_0)^2\epsilon_0 - (1-\epsilon_0) \geq 0.$$

$$\Leftrightarrow (1-\epsilon_0) \left[(1-\epsilon_0)^2 + 3(1-\epsilon_0)\epsilon_0 - 1 \right] \geq 0.$$

$$\Leftrightarrow (1-\epsilon_0) \left[\cancel{1 - 2\epsilon_0 + \epsilon_0^2} + 3\epsilon_0 - \cancel{3\epsilon_0^2} \right] \geq 0.$$

$$\Leftrightarrow (1-\epsilon_0) \left[\epsilon_0 - 2\epsilon_0^2 \right] \geq 0.$$

$$\Leftrightarrow \underbrace{(1-\epsilon_0)}_{\geq 0} \underbrace{\epsilon_0 (1-2\epsilon_0)}_{\geq 0} \geq 0.$$

$$\Leftrightarrow (1-2\epsilon_0) \geq 0.$$

$\Rightarrow$ Improvement possible iff $\epsilon_0 \leq \frac{1}{2}$.

④

②

$$P(A|B) = 1 \Rightarrow \frac{P(A \cap B)}{P(B)} = 1 \Rightarrow P(A \cap B) = P(B).$$

$$\Rightarrow P(B \cap A^c) = 0.$$

$$P(B|A^c) = \frac{P(B \cap A^c)}{P(A^c)} = 0.$$

as $P(B) = P(A \cap B) + P(A^c \cap B).$

False

⑥

$$E[f(Y)g(X)|Y] = \hat{f}(Y) \text{ a function of } Y \text{ s.t.}$$

$$\hat{f}(y) = E[f(y)g(x)|Y=y].$$

$$= \sum_{x \in \mathcal{X}_0} f(y)g(x)P_{X|Y}(x|y)$$

$$= f(y)E[g(x)|Y=y].$$

$$\Rightarrow \hat{f}(Y) = f(Y)E[g(x)|Y].$$

True.

$$(c) \quad \text{Var}(X) = E(\text{Var}(X|Y)) \implies \text{Cov}(X, Y) = 0$$

From total law of variance

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

$$\text{If } \text{Var}(X) = E[\text{Var}(X|Y)]$$

$$\implies \text{Var}(E[X|Y]) = 0.$$

$$\text{Let } f(y) = E[X|Y=y].$$

$$E[X|Y] = f(Y).$$

$$\text{Var}(f(Y)) = 0$$

$$\implies f(Y) = E[f(Y)] \text{ w.p. } 1.$$

$$\implies E[X|Y] = E[E[X|Y]] = E[X].$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= E[E[X|Y]] - E[X]E[Y]$$

using
previous
part.

$$\leftarrow = E[Y E[X|Y]] - E[X]E[Y]$$

$$= E[Y E[X]] - E[X]E[Y]$$

$$= E[X]E[Y] - E[X]E[Y] = 0.$$

(d)

(i) A, B are disjoint $\Rightarrow A, B$ are independent.

$$\Downarrow \\ P(A \cap B) = 0$$

$$\Downarrow \\ P(A \cap B) = P(A)P(B)$$

not true in general.

trivial for the cases where

$$P(A) = 0 \text{ or } P(B) = 0.$$

ii) $P(A \cap B) = P(A)P(B)$ - doesn't imply $P(A \cap B) = 0$.

not true in general.

(iii) $P(A) = P(B) = 0.6$ - ~~A, B~~ A, B are disjoint

not possible as this would imply $P(A \cup B) = 1.2$ 1
contradiction

(iv) $P(A) = P(B) = x$ for some $x \in [0, 1]$.

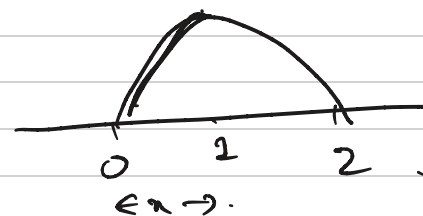
$$P(A \cap B) = x^2 \in [0, 1]$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 2x - x^2$$

$$= x(2 - x).$$

$$\in [0, 1] \quad \forall x \in [0, 1]$$



possible!

5

(a)

A_1 : Event that Alice has head in first trial.

A_2 : " " " " " " second trial

X : # of tosses before the game ends - given that Alice starts the first toss.

Y = # of tosses before the game ends given that Bob starts the first toss
We want to find $E[X]$.

$$E[X] = E[X|A_1] P(A_1) + E[X|A_1^c] P(A_1^c).$$

$$= p E[X|A_1] + (1-p) E[X|A_1^c] \rightarrow \textcircled{1}$$

Game switches to Bob.

$$E[X|A_1] = \underbrace{E[X|A_1 A_2]}_2 \underbrace{P(A_2|A_1)}_p + E[X|A_1 A_2^c] \underbrace{P(A_2^c|A_1)}_{(1-p)}.$$

$$= 2p + (1-p) E[X|A_1 A_2^c] \rightarrow \textcircled{2} \downarrow$$

Game switches to Bob.

$$E[X|A_1^c] = 1 + E[Y]. \rightarrow \textcircled{3}$$

$$E[X | A, A_2^c] = 2 + E[Y] \rightarrow \textcircled{4}$$

$$\text{let } x = E[X] \text{ and } y = E[Y] \rightarrow \textcircled{5}$$

Substituting 2, 3, 4, 5 in ① we have

$$x = p \cdot [ap + (1-p)(2+y)] + (1-p)[1+y]$$

$$= p[2 + (1-p)y] + (1-p) + (1-p)y$$

$$\Rightarrow x = (p+(1-p))y + (1+p)$$

$$\Rightarrow x = (1-p^2)y + (1+p) \rightarrow \textcircled{6}$$

Due to the symmetry by writing similar recursion for $E[Y]$ we get

$$y = (1-q^2)x + (1+q) \rightarrow \textcircled{7}$$

Substituting ⑦ in ⑥ we get

$$x = (1-p^2) \left[(1-q^2)x + (1+q) \right] + (1+p).$$

$$\begin{aligned} \Rightarrow x &= \frac{(1-p^2)(1+q) + (1+p)}{(1-p^2)(1-q^2)} \\ &= \frac{(1+p) \left[(1-p)(1+q) + 1 \right]}{(1-p^2)(1-q^2)}. \end{aligned}$$

⑥ a total of 2 heads.

X_{ij} = # of tosses before the game ends given that Alice starts the toss and Alice has accumulated i heads, Bob j heads.

$$x_{i,j} = E[X_{i,j}].$$

We want to find $x_{0,0}$.

Similarly

$Y_{i,j}$ = # of tosses before the game ends given that Bob starts the toss and Alice has accumulated i heads, Bob j heads.

$$y_{i,j} = E[Y_{i,j}].$$

$$E[X_{0,0}] = p E[X_{0,0} | A_1] + (1-p) E[X_{0,0} | A_1^c].$$

$$E[X_{0,0} | A_1^c] = 1 + E[Y_{0,0}].$$

$$\begin{aligned} E[X_{0,0} | A_1] &= p E[X_{0,0} | A_1, A_2] + (1-p) E[X_{0,0} | A_1, A_2^c] \\ &= p \cdot 2 + (1-p) [2 + E[Y_{1,0}]]. \end{aligned}$$

$$\Rightarrow x_{0,0} = p [2 + (1-p) y_{1,0}] + (1-p)(1 + y_{0,0}). \quad \text{①}$$

By symmetry

$$y_{0,0} = q [2 + (1-q) x_{0,1}] + (1-q) (1 + x_{0,0}) \rightarrow \textcircled{2}$$

$$E[X_{0,1}] = p \underbrace{E[X_{0,1} | A_1]}_{p \times 2 + (1-p) [2 + y_{1,1}]} + (1-p) \underbrace{E[X_{0,1} | A_1^c]}_{(1 + y_{0,1})}$$

$$\Rightarrow x_{0,1} = p (2 + (1-p) y_{1,1}) + (1-p) (1 + y_{0,1}) \rightarrow \textcircled{3}$$

By symmetry

$$y_{1,0} = q (2 + (1-q) x_{1,1}) + (1-q) (1 + x_{1,0}) \rightarrow \textcircled{4}$$

$$E[X_{1,0}] = p \underbrace{E[X_{1,0} | A_1]}_{1 \text{ (Alice has already accumulated a head so game ends)}} + (1-p) \underbrace{E[X_{1,0} | A_1^c]}_{(1 + y_{1,0})}$$

$$\Rightarrow x_{1,0} = p + (1-p) (1 + y_{1,0}) \rightarrow \textcircled{5}$$

Similarly

$$y_{0,1} = q + (1-q)(1+x_{0,1}) \rightarrow \textcircled{6}$$

$$E[x_{1,1}] = p \underbrace{E[x_{11}|A_1]}_1 + (1-p) \underbrace{E[x_{11}|A_1^c]}_{1+y_{1,1}}$$

$$\Rightarrow x_{1,1} = p + (1-p)(1+y_{1,1}) \\ = 1 + (1-p)y_{1,1} \rightarrow \textcircled{7}$$

Similarly

$$y_{1,1} = q + (1-q)(1+x_{1,1}) \\ = 1 + (1-q)x_{1,1} \rightarrow \textcircled{8}$$

$$\Rightarrow x_{1,1} = 1 + (1-p) + (1-p)(1-q)x_{1,1}$$

$$\Rightarrow x_{11} = \frac{2-p}{1-(1-p)(1-q)}$$

$$y_{11} = \frac{2-q}{1-(1-p)(1-q)}$$

⑥. (a) 30 balls, 10 are red, 8 are blue, 12 balls are drawn.

(a) # of Outcomes in which
 i -th red ball gets picked = $(30-1)C_{11}$
 total possible outcomes = $30C_{12}$.

Probability that i -th red ball gets picked =
$$\frac{{}^{29}C_{11}}{{}^{30}C_{12}} = \frac{12}{30} = \frac{2}{5}$$

(b) Prob that i -th red & j -th blue ball get picked =
$$\frac{{}^{28}C_{10}}{{}^{30}C_{12}} = \frac{28!}{10! 18!} \cdot \frac{30!}{12! 18!}$$

=
$$\frac{11 \times 12}{29 \times 30}$$

(c) $E[X]$, $E[Y]$.

$X = X_1 + X_2 + \dots + X_{10}$
 # of red balls.

$$P(X_i = 1) = 1 - P(X_i = 0) = 2/5.$$

X_i is 1 if ith red ball is picked, its 0 otherwise

$$\Rightarrow E[X] = \sum_{i=1}^{10} E[X_i] = 10 \times \frac{2}{5} = 4.$$

$Y_j = 1$ if j blue ball is picked
0 otherwise.

$$P(Y_j = 1) = \frac{{}^{29}C_{11}}{{}^{30}C_{12}} = \frac{12}{30} = 2/5.$$

$$Y = \# \text{ of blue balls picked} \\ = Y_1 + Y_2 + \dots + Y_8$$

$$\Rightarrow E[Y] = 8 \times \frac{2}{5} = \frac{16}{5} = 3.2$$

(d)

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

$$E[XY] = E\left[\sum_{i=1}^{10} x_i \sum_{j=1}^8 y_j\right]$$

$$= E\left[\sum_{i,j} x_i y_j\right]$$

$$= \sum_{i,j} E[x_i y_j]$$

$$E[x_i y_j] = \text{Prob that } i\text{th red ball \& } j\text{th blue ball get picked}$$

$$= \frac{11 \times 12}{29 \times 30}$$

$$\Rightarrow E[X, Y] = 10 \times 8 \times \frac{11 \times 12}{29 \times 30}$$

$$\text{Cov}(X, Y) = \frac{10 \times 8 \times 11 \times 12}{29 \times 30} - (4)(3.2)$$